

CIS41032: Practical for Advance Database Management System
Department of ICT Faculty of Technology
South Eastern University of Sri Lanka

Academic Year: 2020/2021

Lab Sheet 10

Title: Work with Data Import and Aggregation Pipeline.

Objective:

- Import the Dataset to MongoDB.
- Working with Aggregation Framework.

Setup Instructions (Pre-requisite):

- Download and Install MongoDB Tools: <https://www.mongodb.com/try/download/database-tools>
- Download the appropriate version for your OS and install it.

Set Up Environment Variable (Path):

- Go to: System Properties → Environment Variables → Under System Variables → Select Path → Click Edit → Add the path where MongoDB tools (mongoimport) are installed.

C:\Program Files\MongoDB\Database Tools\100\bin

Verify Installation:

- Open CMD and run:

mongoimport --version

Importing Dataset into MongoDB

- Importing a JSON File:

cd C:\Path\To\Your\MongoDB\bin

mongoimport --jsonArray --db mydb --collection mycollection --file cars.json

Exercise:

- Import the dataset into a MongoDB database named CarSalesDB, collection cars.

```
mongoimport --jsonArray --db CarSalesDB --collection cars --file carsales.json
```

- Verify the creation of the database and collection using the following MongoDB shell (mongosh) commands

```
Show dbs
use CarSalesDB
Show Collections
db.cars.find().pretty()
```

- a. Find all cars sold by brand Toyota.
- b. Find all cars with a price greater than 30000.
- c. Return the car_id, brand, and price of all cars sorted by price descending.
- d. Find all Ford cars sold after 2024-07-01.
- e. Count how many cars were sold by each brand. (\$group)
- f. Find the total revenue from car sales.
- g. Project only the brand, model, and sold_date for all documents.
- h. Use \$match and \$group to find the total price of Toyota cars sold.